R-UNIVERSALITY OF THE SPECIAL UNITARY LIE ALGEBRA OF A HERMITIAN FORM OVER A QUATERNION DIVISION ALGEBRA

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Allowed standard prerequisites (editorial).

- The Tits algebra of the natural representation of the unitary algebra is the division algebra D. Used inside the counted proof at source lines 585-586 and again at source lines 591-592, attributed by the author to page 378 of the cited book KnusEtAl-BookOfInvols. Identifies the Tits algebra whose base change decides universality.
- The minimal-extension lemma for the star-action. The author's lemma labelled SameFixed, used inside the counted proof at source line 579. Makes the quadratic extension L imaginary because the algebra is outer.

Source reference. Original lemma 2.3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 471–471. Navigation pointer only: the counted proof uses this lemma once to conclude that the extension is imaginary; its content is declared as a named standard prerequisite and it has its own card in this lane.

Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer only: the counted proof invokes the author's equivalence for its closing steps.

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Source reference. Original definition 4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 432–432. Navigation pointer only: the labelled item of that equivalence used at source line 588.

Proposition 0.1. *Let* $\mathfrak{g} = \mathfrak{su}_n(B; D, \tau)$*, where*

- *D is a division algebra that is central over a quadratic extension L of $\mathbb{Q}$,*
- *τ is an anti-involution of D that is nontrivial on L, and*
- *B is an invertible τ-Hermitian matrix in* $\mathrm{Mat}_{n\times n}(D)$.

Then $\mathfrak{g}$ is $\mathbb{R}$-universal if and only if $D = L$ is an imaginary quadratic extension of $\mathbb{Q}$, and either n is odd or $(-1)^{n/2} \det B$ is either negative or the norm of some element of L.

Proof. We prove only ($\Rightarrow$), but the argument is reversible. Since $\mathfrak{g}$ is outer, Lemma 2.3 implies that $\mathfrak{g} \otimes_{\mathbb{Q}} \mathbb{R}$ is also outer, and the quadratic extension L is imaginary.

Let d be the degree of D over L, and let λ be the highest weight of the standard representation of $\mathfrak{sl}_{dn}(\mathbb{C})$, so $\{\lambda^i\}_{i=1}^{dn}$ is a set of representatives of Z^*.

The Tits algebra of the natural representation $\rho\colon \mathfrak{g} \hookrightarrow \mathrm{Mat}_{n\times n}(D)$ is D [4, p. 378]. Since D splits over $\mathbb{R}$ (recall that the center L is an imaginary extension), we have $D_{\mathfrak{g},\mathbb{R}}(\lambda) = \mathbb{C}$, so we conclude from Lemma 2.1(4) that $D = L$.

If λ is any weight that is not fixed by the $*$-action, then $D_{\mathfrak{g},\mathbb{Q}}(\lambda) = L$ and $D_{\mathfrak{g},\mathbb{R}}(\lambda) = \mathbb{C}$ [4, p. 378], so $D_{\mathfrak{g},\mathbb{Q}}(\lambda) \otimes_{\mathbb{Q}} \mathbb{R} = D_{\mathfrak{g},\mathbb{R}}(\lambda)$, as specified in Lemma 2.1(4). Hence, such weights (which are all of the weights when n is odd) do not place any further restriction on $\mathfrak{g}$.

Suppose n is even. Any weight that is fixed by the $*$-action (and is not in the root lattice) is congruent to $\lambda^{n/2}$, modulo the root lattice. Write $L = \mathbb{Q}\big[\sqrt{a}\big]$ and let $b = (-1)^{n/2} \det B$. Then, for F either $\mathbb{Q}$ or $\mathbb{R}$, $D_{\mathfrak{g},F}(\lambda^{n/2})$ is Brauer equivalent to the quaternion algebra $(a, b)_F$ [4, p. 378 and Corollary 10.35 on p. 131)]. This is trivial in the Brauer group if and only if it is split, which means that b is the norm of some element of $F\big[\sqrt{a}\big]$.

Since $\mathfrak{g}$ is $\mathbb{R}$-universal, we see from Lemma 2.1(4) that either $D_{\mathfrak{g},\mathbb{R}}(\lambda^{n/2}) = \mathbb{H}$ (which, by the preceding paragraph, means that b is not a norm in $\mathbb{R}\big[\sqrt{a}\big] = \mathbb{C}$, so $b < 0$) or $D_{\mathfrak{g},\mathbb{Q}}(\lambda^{n/2}) = \mathbb{Q}$ (which means that b is a norm in $\mathbb{Q}\big[\sqrt{a}\big] = L$). □

References

[1] Demazure M., Grothendieck A., Schémas en groups. III. Structure des schémas en groupes reductifs, Séminaire de géométrie algébrique du Bois Marie, 1962/1964, available at `http://library.msri.org/books/sga/sga/pdf/sga3-3.pdf`.

[2] Harder G., Bericht über neuere Resultate der Galoiskohomologie halbeinfacher Gruppen, *Jber. Deutsch. Math.-Verein.* **70** (1968), 182–216, available at `http://www.digizeitschriften.de/en/dms/img/?PPN=PPN37721857X_0070&DMDID=dmdlog17`.

[3] Kneser M., Lectures on Galois cohomology of classical groups, *Tata Institute of Fundamental Research Lectures on Mathematics*, Vol. 47, Tata Institute of Fundamental Research, Bombay, 1969, available at `http://www.math.tifr.res.in/~publ/ln/tifr47.pdf`.

[4] Knus M.A., Merkurjev A., Rost M., Tignol J.P., The book of involutions, *American Mathematical Society Colloquium Publications*, Vol. 44, Amer. Math. Soc., Providence, RI, 1998.

[5] Lam T.Y., Introduction to quadratic forms over fields, *Graduate Studies in Mathematics*, Vol. 67, Amer. Math. Soc., Providence, RI, 2005.

[6] Morris D., Real representations of semisimple Lie algebras have $\mathbb{Q}$-forms, in Algebraic Groups and Arithmetic, Tata Institute of Fundamental Research, Mumbai, 2004, 469–490.

[7] Platonov V., Rapinchuk A., Algebraic groups and number theory, *Pure and Applied Mathematics*, Vol. 139, Academic Press, Inc., Boston, MA, 1994.

[8] Prasad G., Rapinchuk A., On the existence of isotropic forms of semi-simple algebraic groups over number fields with prescribed local behavior, *Adv. Math.* **207** (2006), 646–660.

[9] Raghunathan M.S., Arithmetic lattices in semisimple groups, *Proc. Indian Acad. Sci. Math. Sci.* **91** (1982), 133–138.

[10] Tits J., Classification of algebraic semisimple groups, in Algebraic Groups and Discontinuous Subgroups (Proc. Sympos. Pure Math., Boulder, Colo., 1965), Amer. Math. Soc., Providence, R.I., 1966, 33–62.
[11] Tits J., Représentations linéaires irréductibles d'un groupe réductif sur un corps quelconque, *J. Reine Angew. Math.* **247** (1971), 196–220.
[12] Tits J., Strongly inner anisotropic forms of simple algebraic groups, *J. Algebra* **131** (1990), 648–677.